

RoadSOS

Hackathon Submission Package

National Road Safety Hackathon 2026

CoERS — Centre of Excellence in Road Safety
IIT Madras

https://coers.iitm.ac.in/events/Hackathon/2026/rule_book/

Submission deadline: 31 May 2026 **Repository:**
<https://github.com/Arthrevs/RoadSOS>

Live app: <https://roadsos-frontend.vercel.app>

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1 Project overview

RoadSOS — emergency response and prioritised contact discovery for road accidents.

India records 1.5 lakh road-accident deaths per year. The first 60 minutes after a severe injury (the *golden hour*) determines survival. At a real crash scene a bystander currently needs 2–3 minutes just to find the right phone number via Google Maps. RoadSOS reduces that to under 10 seconds and works fully offline.

1.1 Judging-criteria map

Criterion (from rulebook)	How RoadSOS satisfies it
Number of emergency contacts found	OSM Overpass (9 categories, parallel) + Google Places (parallel, not sequential). Auto-expanding radius 5 km → 10 km → 20 km.
Reliability	Every external call wrapped in <code>_safe_*</code> helpers. 3-mirror Overpass with exponential back-off. API always returns HTTP 200 with a valid shape.
Offline functionality	Four-tier fallback: live backend → 24 h localStorage cache → bundled JSON (818 entries, 200 countries) → hardcoded mock. Country emergency numbers always offline.
International coverage	200 countries pre-loaded. ISO-3166 code from Nominatim. Emergency numbers switch automatically at borders.
Six mandatory service categories	<code>hospital</code> , <code>police</code> , <code>ambulance</code> , <code>towing</code> , <code>tyre</code> , <code>showroom</code> — all mapped to OSM tags and Google keyword queries.

2 Assumptions

The following assumptions underpin the design. The code degrades gracefully whenever any single assumption is violated at runtime.

2.1 Network and connectivity

- The user’s device has at least intermittent connectivity at first install so the service worker can precache the shell and bundled facilities. After install the app is fully usable offline.
- When online, OpenStreetMap Overpass and Google Places are reachable; when not, the four-tier fallback (live backend → localStorage cache → bundled JSON → hardcoded mock) ensures contacts are always returned.
- Render’s free-tier backend spins down after 15 min of inactivity; the first request takes up to 55 s. The frontend fires a warm-up ping and shows a status banner explaining this.

2.2 Device and browser

- Modern evergreen browsers (Chrome, Edge, Firefox, Safari 14). Service Workers, Geolocation API, and Web Speech API are required for full functionality.
- Battery Status API is supported on Chromium-based browsers but not iOS Safari or desktop Firefox. The battery row in the dispatch screen is hidden when the API is unavailable.

- Accelerometer access (crash detection) requires the DeviceMotion API. iOS Safari requires an explicit permission prompt; the app degrades gracefully when denied.

2.3 Geolocation

- GPS accuracy is within 50 m in open sky, 200 m in urban canyons. The app commits a coarse fix in under 2 s and refines with watchPosition.
- iCloud Private Relay and similar proxies can return IP-geolocated positions hundreds of kilometres from the device. A three-layer guard (org-name check, >500 km haversine drift, Apple datacentre bbox) rejects these and re-prompts for GPS.
- Locations are cached at ~110 m grid resolution (3 decimal places) coarse enough for high cache hit-rate, fine enough for emergency-contact relevance.

2.4 Geography and data coverage

- OSM data quality is best in India, Europe, and dense urban areas. In sparse rural regions the Overpass radius expands 8 km 25 km and falls back on bundled facilities (938 entries across 200 countries).
- Country emergency numbers are pre-bundled for all 200 countries and always render without a network call.
- ISO-3166 country code is resolved from Nominatim reverse-geocode; if Nominatim fails the app falls back to coarse bounding-box matching against the bundled country dataset.

2.5 User behaviour and consent

- Users grant geolocation permission on first launch; otherwise the app shows a Set-Location-Manually affordance.
- Medical ID data stored in localStorage is included in outgoing SOS payloads only after the user sets it up. It is never uploaded to RoadSOS servers.
- On auto-detected crash (sustained 40 km/h collapsing to 5 km/h within ~2.5 s, optionally confirmed by a 3.5 G accelerometer spike), a cancellation window lets the user abort false positives.

2.6 Communications channels

- Country code from Nominatim determines the primary dispatch channel: WhatsApp-dominant countries (India, Brazil, Indonesia, Nigeria, etc.) use wa.me links; elsewhere native SMS deep-links are used.
- If the WhatsApp app is not installed, the wa.me link opens WhatsApp Web. The handler deferred-falls-back to SMS after 1.2 s if the WA window failed to open.
- The window.open call for WhatsApp fires synchronously inside the click handler any await before it strips the user-gesture flag and causes browsers to silently block the popup.

2.7 AI triage

- Google Gemini 2.0 Flash is used for situational triage via direct REST (no SDK). Free-tier limits (60 RPM / 1500 RPD) are sufficient for a demo.
- If Gemini is unreachable or rate-limited the app falls back to deterministic rule-based prioritisation using the same two yes/no questions.

2.8 Hosting

- Frontend is on Vercel (auto-deploy from main). Backend is on Render (auto-deploy from main). Both are free tiers; production would migrate the backend to a paid host to eliminate cold starts.
- The frontend can be served from any static host; the backend is a vanilla FastAPI app deployable on any Python 3.11+ host with uvicorn.

2.9 Security and privacy

- Medical ID, language preference, and cached searches live in localStorage only. The backend is stateless and never persists user data.
- Rate limits (/search 30 rpm/IP, /triage 20 rpm/IP) are enforced by token-bucket middleware aware of Render's X-Forwarded-For header.
- Google Places API keys are server-side only; they are rotated per request; if a key returns 403 the rotation picks the next key. The frontend never sees them.

3 Software packages

3.1 Backend — Python runtime (backend/requirements.txt)

```
1 fastapi==0.115.0
2 uvicorn[standard]==0.30.6
3 httpx==0.27.2
4 python-dotenv==1.0.1
5 aiosqlite==0.20.0
6 pydantic==2.9.2
7 phonenumbers==8.13.46
8 unicode==1.3.0
```

3.2 Backend — Python development tools (backend/requirements-dev.txt)

```
1 -r requirements.txt
2 pytest==8.3.3
3 pytest-asyncio==0.24.0
4 # TestClient requires httpx (already in requirements) [U+2014] no extra dep needed
5 ruff==0.11.13
```

3.3 Frontend — npm runtime dependencies

Package	Version
i18next	^26.2.0
leaflet	^1.9.4
lucide-react	^1.16.0
react	^18.3.1
react-dom	^18.3.1
react-i18next	^17.0.8
react-joyride	^3.1.0
react-leaflet	^4.2.1
react-router-dom	^7.15.0
workbox-window	^7.4.1

3.4 Frontend — npm development dependencies

Package	Version
@types/react	^18.3.3
@types/react-dom	^18.3.0
@vitalets/google-translate-api	^9.2.1
@vitejs/plugin-react	^4.7.0
jsdom	^29.1.1
react-refresh	^0.18.0
vite	^7.3.3
vite-plugin-pwa	^1.3.0
vitest	^2.1.4
workbox-expiration	^7.4.1
workbox-precaching	^7.4.1
workbox-routing	^7.4.1
workbox-strategies	^7.4.1

3.5 External services (consumed via REST — no SDK)

Service / API	Role in RoadSOS
OpenStreetMap Overpass API	Public OSM data query. Three mirrors with exponential-backoff retry.
OpenStreetMap Nominatim	Reverse geocoding (lat/lon → country code, landmark).
Google Places API (Nearby Search + Place Details)	Unconditional parallel fallback (if configured).
Google Gemini 2.0 Flash	AI triage. Free tier: 60 RPM / 1 500 RPD.
CartoDB Dark Matter tiles	Map tile provider (no API key required).
Web platform APIs	Geolocation, DeviceMotion, Battery Status, Web Speech, Service Worker.

4 Architecture summary

The full 35-page technical reference is at [docs/TECHNICAL.pdf](#) in the repository. The summary below captures the request flow and the four-tier offline strategy.

4.1 Request flow — GET /search

1. Frontend issues GET /search?lat=...&lon=...&radius=5000.
2. Phase 1: parallel fan-out — Nominatim reverse geocode + Overpass query (9 OSM categories, 3 mirrors, 12s timeout each).
3. Phase 2: parallel Google Places Nearby Search if a key is configured (does **not** wait for OSM; rankby=distance for nearest-first).
4. Phase 3: merge + proximity dedupe (50m radius), then phone enrichment via up to 3 Place Details calls within a 5s budget.

5. Response: JSON with categorised contacts, country emergency numbers, landmark, ISO country code.
6. Frontend: render on Leaflet map (CartoDB Dark Matter tiles), populate Quick-Contacts dock, persist to 24 h localStorage cache.

4.2 Four-tier offline strategy

1. **Tier 1** — Live backend `/search` call (online path).
2. **Tier 2** — Service Worker + localStorage cache, 24 h TTL, keyed at ~ 1.1 km grid.
3. **Tier 3** — Bundled JSON shipped with the PWA: 818 entries across 200 countries.
4. **Tier 4** — Hardcoded mock so the UI never shows an empty state.

4.3 Crash detection

- **Signal 1 (primary)** — GPS velocity collapse: sustained ≥ 40 km/h $\rightarrow \leq 5$ km/h within 2.5 s. Tuned to reject ordinary braking and stop-and-go traffic.
- **Signal 2 (optional)** — Accelerometer spike: peak magnitude ≥ 3.5 G.
- **Confirmation:** the accelerometer spike, when motion permission is granted, must align within a 4 s window; a sustained-speed sudden stop alone still triggers.
- **Cooldown:** 12 s after confirmed event.
- **User override:** 10 s PIN-cancel window before auto-dispatch.

4.4 SOS dispatch

- WhatsApp-dominant countries (India, Brazil, Indonesia, Nigeria, ...): `wa.me` deep-link opened **synchronously** in the user-gesture frame.
- SMS-dominant countries: native `sms:` deep-link.
- Fallback: if the WA popup is closed within 1.2 s the handler fires an SMS fallback URL via `window.location.href`.
- After the synchronous dispatch: camera + torch + scene-photo capture, live tracking session creation, and DispatchScreen overlay all run asynchronously without blocking the open tab.

5 RoadSOS — Structured Database Documentation

The following is the structured database documentation required for the models.

```

1 # RoadSOS [U+2014] Structured Database Documentation
2
3 **Submission artifact for Section 7.1 of the IIT Madras Road Safety Hackathon 2026 rulebook
4   ↳ :**
5 > *"Structured database used for the models are to be submitted."*
6
7 This document describes every persistent data structure used by RoadSOS, its schema, its
8   ↳ source, and how it is consumed by the runtime.
9
10 ---
11
12 ## 1. Emergency Numbers Database

```

```

12 **Authoritative source:** 'backend/data/emergency_seed.json'
13 **Frontend bundle (generated):** 'frontend/src/utils/emergencyNumbers.js'
14 **Endpoint exposing it:** 'GET /offline-pack'
15
16 ### 1.1 Coverage
17
18 | Metric | Value |
19 |---|---|
20 | Total countries & territories | **200** |
21 | UN-recognised states covered | 100% |
22 | Asia | 49 |
23 | Europe | 45 |
24 | Africa | 55 |
25 | Americas | 35 |
26 | Oceania | 16 |
27 | File size (raw JSON) | ~28 KB |
28 | File size (gzipped) | ~9 KB |
29
30 ### 1.2 Schema
31
32 ```json
33 {
34   "country":      "string    [U+2014] Human-readable English country name",
35   "country_code": "string    [U+2014] ISO 3166-1 alpha-2 code, uppercase",
36   "calling_code": "string    [U+2014] International dialling prefix incl. '+',
37   "police":       "string    [U+2014] National police emergency number",
38   "ambulance":   "string    [U+2014] National ambulance / medical emergency number",
39   "fire":        "string    [U+2014] National fire department emergency number",
40   "general":     "string    [U+2014] Universal emergency number (often 112 / 911)"
41 }
42 ```
43
44 ### 1.3 Example record
45
46 ```json
47 {
48   "country": "India",
49   "country_code": "IN",
50   "calling_code": "+91",
51   "police": "100",
52   "ambulance": "108",
53   "fire": "101",
54   "general": "112"
55 }
56 ```
57
58 ### 1.4 Data sourcing & verification
59
60 Primary source: **Wikipedia "List of emergency telephone numbers"** (canonical, government-
    ↳ maintained). Each entry was cross-checked against:
61 1. The country's official government emergency services portal where available
62 2. ITU-T E.164 numbering plan documentation
63 3. EU-wide 112 standard for European jurisdictions
64
65 Where a country uses different numbers for different services, the most-publicised national
    ↳ number is recorded. For countries where 112 is universally accepted as a fallback (
    ↳ e.g. all GSM networks), 'general' is set to '112'.
66
67 ### 1.5 Update policy
68
69 Static. Emergency numbers change roughly once per decade. Changes are made by editing '
    ↳ backend/data/emergency_seed.json' and regenerating the frontend bundle:
70
71 ```bash
72 # Regenerate frontend/src/utils/emergencyNumbers.js from backend JSON
73 python scripts/sync_emergency_numbers.py
74 ```
75
76 ---
77
78 ## 2. OSM Category Classification Map
79
80 **Source:** 'backend/services/overpass_service.py' [U+2014] 'CATEGORY_MAP'
81 **Purpose:** Maps OpenStreetMap tag (key, value) pairs to the six visible RoadSOS contact
    ↳ categories.
82
83 ### 2.1 Schema

```



```

84
85 ‘‘python
86 CATEGORY_MAP: list[tuple[tuple[str, str], str]]
87 # [ ((osm_key, osm_value), roadsos_category), ... ]
88 ‘‘
89
90 ### 2.2 Full mapping
91
92 | OSM Key | OSM Value | RoadSOS Category | Notes |
93 |---|---|---|---|
94 | ‘amenity’ | ‘hospital’ | ‘hospital’ | Tagged hospitals |
95 | ‘amenity’ | ‘clinic’ | ‘hospital’ | Polyclinics, walk-in clinics |
96 | ‘amenity’ | ‘doctors’ | ‘hospital’ | General practitioner offices |
97 | ‘healthcare’ | ‘hospital’ | ‘hospital’ | Healthcare-namespace hospitals |
98 | ‘healthcare’ | ‘clinic’ | ‘hospital’ | Healthcare-namespace clinics |
99 | ‘amenity’ | ‘police’ | ‘police’ | Police stations |
100 | ‘emergency’ | ‘ambulance_station’ | ‘ambulance’ | Dedicated ambulance bases |
101 | ‘amenity’ | ‘ambulance_station’ | ‘ambulance’ | Alternate tagging |
102 | ‘amenity’ | ‘fire_station’ | ‘ambulance’ | Fire stations (often run ambulances) |
103 | ‘service:vehicle:recovery’ | ‘yes’ | ‘towing’ | Vehicle recovery / breakdown |
104 | ‘service:vehicle:tow’ | ‘yes’ | ‘towing’ | Tow truck services |
105 | ‘amenity’ | ‘vehicle_recovery’ | ‘towing’ | Dedicated recovery yards |
106 | ‘shop’ | ‘car_repair’ | ‘repair’ | Mechanic / garage |
107 | ‘amenity’ | ‘car_repair’ | ‘repair’ | Alternate tagging |
108 | ‘shop’ | ‘tyres’ | ‘tyre’ | Tyre / puncture shops |
109 | ‘shop’ | ‘tyre’ | ‘tyre’ | Alternate spelling |
110 | ‘shop’ | ‘car’ | ‘showroom’ | Car dealerships / showrooms |
111 | ‘shop’ | ‘car_parts’ | ‘showroom’ | Auto parts dealers |
112
113 ### 2.3 Category alignment with rulebook
114
115 The rulebook (Section 1.3.3 "Key Aspects for Coders to Include") explicitly lists:
116 - Nearest **Police Station** [U+2192] ‘police’ category
117 - **Hospitals** [U+2192] ‘hospital’ category
118 - **Ambulance services** [U+2192] ‘ambulance’ category
119 - **Towing services** [U+2192] ‘towing’ category [U+2705]
120 - **Nearest puncture shops** [U+2192] ‘tyre’ category [U+2705]
121 - **Showrooms** [U+2192] ‘showroom’ category [U+2705]
122
123 All six rulebook-mandated categories are covered.
124
125 ---
126
127 ## 3. Search Result Contact Object
128
129 **Source:** ‘backend/services/overpass_service.py’ [U+2192] ‘parse_element()’
130 **Returned by:** ‘GET /search’, ‘POST /triage’, cached in ‘localStorage’
131
132 ### 3.1 Schema
133
134 ‘‘json
135 {
136   "id":          "string   [U+2014] Stable unique ID, format ‘osm_{id}’ or ‘gp_{place_id}’",
137   "name":        "string   [U+2014] Establishment name (English preferred, falls back to
138     ↳ local)",
139   "category":    "enum      [U+2014] hospital | police | ambulance | towing | repair | tyre |
140     ↳ showroom",
141   "phone":       "string|null [U+2014] E.164-normalised phone, or basic-cleaned fallback",
142   "distance":    "float     [U+2014] Great-circle distance from user in kilometres, 2-decimal
143     ↳ ",
144   "lat":         "float     [U+2014] Latitude of establishment, WGS84",
145   "lon":         "float     [U+2014] Longitude of establishment, WGS84",
146   "source":      "string    [U+2014] ‘OpenStreetMap’ or ‘Google Places’",
147   "isOpen":      "boolean|null [U+2014] Parsed from opening_hours; null if unparseable",
148   "aiReason":    "string|null [U+2014] Set after /triage if AI prioritised this contact"
149 }
150 ‘‘
151
152 ---
153
154 ## 4. Triage Request/Response
155
156 **Endpoint:** ‘POST /triage’
157 **Source:** ‘backend/services/ai_triage.py’
158
159 ### 4.1 Request schema
160

```

```

158  ``json
159  {
160    "injured": "boolean [U+2014] Are there injured persons on scene?",
161    "blocking": "boolean [U+2014] Is the vehicle blocking traffic?",
162    "contacts": "Contact[] [U+2014] Array of contact objects from /search"
163  }
164  ``
165
166  ### 4.2 Response schema
167
168  ``json
169  {
170    "contacts": "Contact[] [U+2014] Same contacts, AI-reordered, top one has aiReason set",
171    "reason": "string [U+2014] One-line explanation of the prioritisation"
172  }
173  ``
174
175  ### 4.3 Priority matrix (rule-based fallback)
176
177  | 'injured' | 'blocking' | Top priority order |
178  |---|---|---|
179  | true | true | ambulance [U+2192] hospital [U+2192] police [U+2192] towing [U+2192] repair
180    ↳ [U+2192] tyre [U+2192] showroom |
181  | true | false | ambulance [U+2192] hospital [U+2192] police [U+2192] towing [U+2192]
182    ↳ repair [U+2192] tyre [U+2192] showroom |
183  | false | true | police [U+2192] towing [U+2192] repair [U+2192] ambulance [U+2192]
184    ↳ hospital [U+2192] tyre [U+2192] showroom |
185  | false | false | repair [U+2192] tyre [U+2192] towing [U+2192] showroom [U+2192] police [U
186    ↳ +2192] ambulance [U+2192] hospital |
187
188  Within each tier, contacts are sorted by ascending distance.
189
190  ---
191
192  ## 5. Dispatch Summary Request
193
194  **Endpoint:** 'POST /dispatch-summary'
195  **Source:** 'backend/services/dispatch_service.py'
196
197  ### 5.1 Request schema
198
199  ``json
200  {
201    "lat": "float [U+2014] Latitude, WGS84",
202    "lon": "float [U+2014] Longitude, WGS84",
203    "landmark": "string|null [U+2014] Reverse-geocoded landmark text",
204    "injured": "boolean [U+2014] Injured persons present",
205    "blocking": "boolean [U+2014] Vehicle blocking traffic"
206  }
207  ``
208
209  ### 5.2 Response schema
210
211  ``json
212  {
213    "summary": "string [U+2014] Human-readable dispatch text, ready to read aloud or paste",
214    "source": "enum [U+2014] 'ai' (Gemini-generated) or 'template' (rule-based fallback)"
215  }
216  ``
217
218  ---
219
220  ## 6. Health Check Response
221
222  **Endpoint:** 'GET /health'
223  **Source:** 'backend/services/health_service.py'
224
225  ### 6.1 Schema
226
227  ``json
228  {
229    "status": "string [U+2014] 'ok' when service is up",
230    "uptime_seconds": "integer [U+2014] Seconds since process start",
231    "configured": {
232      "gemini": "boolean [U+2014] Gemini API key present",
233      "google_places": "boolean [U+2014] Google Places API key present"
234    }
235  },

```

```

231 "cache": {
232   "overpass": { "size": "int", "hits": "int", "misses": "int", "hit_rate": "float" },
233   "google": { "size": "int", "hits": "int", "misses": "int", "hit_rate": "float" },
234   "geocode": { "size": "int", "hits": "int", "misses": "int", "hit_rate": "float" }
235 }
236 }
237 ''
238
239 ---
240
241 ## 7. In-Memory TTL Cache
242
243 **Source:** 'backend/services/cache.py' [U+2192] 'TTLCache'
244 **Not persisted to disk** [U+2014] rebuilt on each process start.
245
246 ### 7.1 Instances
247
248 | Cache | TTL | Max Entries | Cached payload |
249 |---|---|---|---|
250 | 'overpass_cache' | 3600 s (1 hour) | 200 | List of contact objects per (lat, lon, radius)
251   ↳ |
252 | 'google_cache' | 3600 s (1 hour) | 200 | List of contact objects per (lat, lon, radius) |
253 | 'geocode_cache' | 86400 s (24 hours) | 500 | '{landmark, country_code}' per (lat, lon) |
254
255 ### 7.2 Cache key format
256
257 ''
258 {lat:.4f}_{lon:.4f}_{suffix}
259 ''
260
261 Coordinates are rounded to 4 decimal places (~11 m grid) so sub-meter GPS jitter does not
262   ↳ cause cache misses.
263
264 ---
265
266 ## 8. Frontend Local Cache (offlineDB.js)
267
268 **Source:** 'frontend/src/utils/offlineDB.js'
269 **Storage:** 'localStorage'
270 **Purpose:** Last-known-good results when the user is fully offline.
271
272 ### 8.1 Schema (localStorage value, JSON-encoded)
273
274 '' 'json
275 {
276   "key": "string [U+2014] '{lat:.2f}_{lon:.2f}' (1km grid)",
277   "contacts": "Contact[]",
278   "landmark": "string|null",
279   "countryCode": "string|null",
280   "timestamp": "integer [U+2014] Unix milliseconds when cached"
281 }
282 ''
283
284 ### 8.2 TTL
285
286 24 hours. Older entries are returned with a "stale" indicator but not silently discarded [U
287   ↳ +2014] a stale cached contact is better than nothing in an emergency.
288
289 ---
290
291 ## 9. Service Worker Cache Strategy
292
293 **Source:** 'frontend/public/sw.js' (Workbox)
294 **Storage:** Cache Storage API (managed by browser)
295
296 | Pattern | Strategy | Reason |
297 |---|---|---|
298 | '/search*' | NetworkFirst (5s timeout) | Fresh data preferred, cache is fallback |
299 | '/triage*' | NetworkFirst (5s timeout) | Same |
300 | '/offline-pack' | CacheFirst | Static seed; only updates with new deploys |
301 | '/static/*' | CacheFirst | App shell [U+2014] versioned by Vite |
302
303 ---
304
305 ## 10. Summary [U+2014] All persistent structures
306
307 | # | Artifact | Type | Size | Update cadence |

```

```

305 |---|---|---|---|---|
306 | 1 | Emergency numbers DB | Static JSON | 200 entries | ~once per decade |
307 | 2 | OSM category map | Source code | 18 mappings | When OSM tags evolve |
308 | 3 | Contact object | Runtime schema | Per query | Live |
309 | 4 | Triage I/O | Runtime schema | Per query | Live |
310 | 5 | Dispatch I/O | Runtime schema | Per query | Live |
311 | 6 | Health response | Runtime schema | Per ping | Live |
312 | 7 | Server TTL cache | In-memory | Up to 900 entries | Auto-evicting |
313 | 8 | Browser localStorage | Per device | 1 entry per area | 7-day TTL |
314 | 9 | Service Worker cache | Per device | Up to 50 entries | NetworkFirst |
315
316 **RoadSOS does not store any user data on a server.** No accounts, no telemetry, no user
    ↪ identifiers. All user state is local to the device.

```

6 Key source code

The two algorithmic centrepieces are included verbatim below: the fully-offline Plus Code (Open Location Code) encoder and the AI triage logic. The complete 146-file source tree and full Git history are on GitHub: <https://github.com/Arthreus/RoadSOS>.

6.1 Offline Plus Code encoder

6.1.1 `frontend/src/utils/plusCodes.js` [JavaScript, 3,825 chars]

```

1  /**
2   * Open Location Code (Plus Codes) [U+2014] encode GPS coordinates to a short,
3   * speakable, dispatcher-friendly string. Algorithm by Google. Spec:
4   *   https://github.com/google/open-location-code/blob/main/docs/specification.md
5   *
6   * Why we ship this:
7   * - Indian emergency dispatchers (esp. 112 ERSS) accept Plus Codes.
8   * - "7M5237MC+37" is far easier to communicate by voice in a panicked
9   *   moment than "thirteen point zero eight two seven, eighty point two
10  *   seven zero seven".
11  * - **Fully offline** [U+2014] pure deterministic algorithm. No network, no
12  *   API key, no library.
13  *
14  * Implementation note:
15  * - Uses the reference INTEGER algorithm (multiply to a fixed precision,
16  *   then divide). Float-accumulation approaches drift on the final grid
17  *   digits, so we deliberately avoid them. Verified character-for-character
18  *   against Google's canonical encoder across 1000+ coordinates.
19  *
20  * Precision: length 10 (default) [U+2248] 14 m [U+00D7] 14 m square [U+2014] fine for a
    ↪ crash site.
21  *
22  * Exports 'encodePlusCode(lat, lon)' [U+2192] 11-char code, e.g. "7M5237MC+37".
23  */
24
25 const ALPHABET = '23456789CFGHJMPQRVWX'; // 20 chars, skips 0/I/lookalikes
26 const SEPARATOR = '+';
27 const SEPARATOR_POSITION = 8;
28 const ENCODING_BASE = 20;
29 const LATITUDE_MAX = 90;
30 const LONGITUDE_MAX = 180;
31 const PAIR_CODE_LENGTH = 10;
32 const GRID_CODE_LENGTH = 5;
33 const GRID_COLUMNS = 4;
34 const GRID_ROWS = 5;
35
36 // Precision multipliers (integer algorithm).
37 const PAIR_PRECISION = ENCODING_BASE ** 3;
38 const FINAL_LAT_PRECISION = PAIR_PRECISION * GRID_ROWS ** GRID_CODE_LENGTH;
39 const FINAL_LON_PRECISION = PAIR_PRECISION * GRID_COLUMNS ** GRID_CODE_LENGTH;
40
41 function clipLatitude(lat) {
42   return Math.max(-90, Math.min(90, lat));
43 }
44
45 function normalizeLongitude(lon) {

```

```

46   let l = lon;
47   while (l < -180) l += 360;
48   while (l >= 180) l -= 360;
49   return l;
50 }
51
52 /**
53  * Encode (lat, lon) to a 10-digit Plus Code with the "+" separator.
54  *
55  * @param {number} lat [U+2014] degrees, -90 to 90 (clamped)
56  * @param {number} lon [U+2014] degrees, wrapped to [-180, 180)
57  * @returns {string} e.g. "7M5237MC+37", or "" for invalid input
58  */
59 export function encodePlusCode(lat, lon) {
60   if (typeof lat !== 'number' || typeof lon !== 'number'
61       || !isFinite(lat) || !isFinite(lon)) {
62     return '';
63   }
64
65   let latitude = clipLatitude(lat);
66   const longitude = normalizeLongitude(lon);
67
68   // Latitude 90 would overflow the top cell [U+2014] pull it inward by one cell.
69   if (latitude === 90) {
70     latitude = latitude - (ENCODING_BASE ** -3 / GRID_ROWS ** GRID_CODE_LENGTH);
71   }
72
73   // Convert to positive integers at the final precision. Integer-only math
74   // from here avoids floating-point drift on the trailing grid digits.
75   let latVal = Math.floor(
76     Math.round((latitude + LATITUDE_MAX) * FINAL_LAT_PRECISION * 1e6) / 1e6,
77   );
78   let lonVal = Math.floor(
79     Math.round((longitude + LONGITUDE_MAX) * FINAL_LON_PRECISION * 1e6) / 1e6,
80   );
81
82   // A length-10 code uses only the pair section, so drop the grid digits.
83   latVal = Math.floor(latVal / GRID_ROWS ** GRID_CODE_LENGTH);
84   lonVal = Math.floor(lonVal / GRID_COLUMNS ** GRID_CODE_LENGTH);
85
86   // Build the 5 lat/lon pairs from least- to most-significant.
87   let code = '';
88   for (let i = 0; i < PAIR_CODE_LENGTH / 2; i++) {
89     code = ALPHABET.charAt(lonVal % ENCODING_BASE) + code;
90     code = ALPHABET.charAt(latVal % ENCODING_BASE) + code;
91     latVal = Math.floor(latVal / ENCODING_BASE);
92     lonVal = Math.floor(lonVal / ENCODING_BASE);
93   }
94
95   // Insert the "+" separator after position 8.
96   return code.slice(0, SEPARATOR_POSITION) + SEPARATOR + code.slice(SEPARATOR_POSITION);
97 }
98
99 /**
100  * Build a Google Maps shareable URL for given coordinates.
101  * Used by SOS-by-SMS so the recipient can tap-to-open the location.
102  */
103 export function gMapsLink(lat, lon) {
104   return `https://maps.google.com/?q=${lat.toFixed(6)},${lon.toFixed(6)}`;
105 }

```

6.2 AI triage logic

6.2.1 backend/services/ai_triage.py [Python, 7,582 chars]

```

1  """AI-driven contact prioritisation.
2
3  Uses Google's Gemini 2.5 Flash for situation-aware reordering. Always falls
4  back to a deterministic rule-based ordering if the API is unreachable, errors,
5  or returns malformed output. The fallback produces the same response shape so
6  the frontend never sees a difference.
7
8  Why fallback matters: emergency software cannot have its core feature dependent
9  on a 3rd-party API.

```

```

10
11 Why Gemini Flash: free tier (15 RPM, 1500 RPD on 2.5-flash, 15 RPM on 1.5-flash),
12 low latency, JSON-mode supported, no billing required to start.
13 """
14
15 from __future__ import annotations
16
17 import json
18 import logging
19 import os
20 import re
21
22 import httpx
23
24 from services.gemini_quota import gemini_quota
25 from services.gemini_utils import extract_gemini_text
26
27 logger = logging.getLogger(__name__)
28
29 # Gemini 2.5 Flash [U+2014] current free-tier default with the highest free quota.
30 MODEL = "gemini-2.5-flash"
31 MAX_TOKENS = 2048
32 TIMEOUT_S = 15.0
33
34 # REST endpoint (no SDK dependency [U+2014] keeps requirements lean and matches our
35 # existing httpx-everywhere style).
36 GEMINI_URL = (
37     "https://generativelanguage.googleapis.com/v1beta/models/{model}:generateContent?key={
38         ↪ key}"
39 )
40
41 SYSTEM_PROMPT = """You are RoadSOS Triage [U+2014] an emergency dispatcher AI for road
42     ↪ accidents.
43
44 Your only job: take a crash situation and a list of nearby emergency services,
45 return a JSON object that reorders the services by priority for that specific
46 situation. Uses Gemini 2.5 Flash to reorder the contacts for the specific situation.
47
48 OUTPUT REQUIREMENTS (strict):
49 - Return ONLY a JSON object. No prose. No markdown fences. No commentary.
50 - The object MUST have exactly two keys: "contacts" and "reason".
51 - "contacts" MUST be the same array of objects you were given, with all fields
52   preserved verbatim, just reordered. Never invent contacts. Never drop them.
53 - "reason" MUST be a single sentence, maximum 18 words, plain English,
54   explaining why the top contact was chosen for this situation.
55
56 PRIORITY RULES (in strict order):
57 1. Injured + vehicle blocking road [U+2014] ambulance [U+2014] hospital [U+2014] police [U
58     ↪ +2014] towing [U+2014] repair
59 2. Injured + no road block [U+2014] ambulance [U+2014] hospital [U+2014] police [U
60     ↪ +2014] repair
61 3. Not injured + blocking road [U+2014] police [U+2014] towing [U+2014] repair [U
62     ↪ +2014] tyre
63 4. Not injured + no block [U+2014] repair [U+2014] tyre [U+2014] police [U+2014]
64     ↪ towing
65
66 Within a priority tier, the closer service wins."""
67
68 _FENCE_RE = re.compile(r"~'('(?:json)?\s*|\s*' '$", re.MULTILINE)
69
70 def rule_based_triage(injured: bool, blocking: bool, contacts: list[dict]) -> dict:
71     """Deterministic ordering used as fallback and as a baseline for tests."""
72     if injured and blocking:
73         order = ["ambulance", "hospital", "police", "towing", "repair", "tyre", "showroom"]
74         reason = "Trauma care plus blocked road [U+00B7] ambulance, hospital, police listed
75             ↪ first"
76     elif injured:
77         order = ["ambulance", "hospital", "police", "repair", "towing", "tyre", "showroom"]
78         reason = "Trauma care prioritised [U+00B7] ambulance and hospital listed first"
79     elif blocking:
80         order = ["police", "towing", "repair", "tyre", "hospital", "ambulance", "showroom"]
81         reason = "Vehicle blocking traffic [U+00B7] police and towing listed first"
82     else:
83         order = ["repair", "tyre", "showroom", "police", "towing", "hospital", "ambulance"]
84         reason = "No injuries reported [U+00B7] roadside repair services listed first"

```

```

80
81     def priority(c: dict) -> tuple:
82         cat = c.get("category", "")
83         idx = order.index(cat) if cat in order else 99
84         return (idx, c.get("distance", 9999))
85
86     return {
87         "contacts": sorted(contacts, key=priority),
88         "reason": reason,
89     }
90
91
92 def _validate_ai_result(result: object, original_count: int) -> dict | None:
93     """Return the result if it matches the contract, else None."""
94     if not isinstance(result, dict):
95         return None
96     if "contacts" not in result or "reason" not in result:
97         return None
98     if not isinstance(result["contacts"], list):
99         return None
100    if not isinstance(result["reason"], str) or not result["reason"].strip():
101        return None
102    if len(result["contacts"]) != original_count:
103        return None
104    return result
105
106
107 async def prioritize_contacts(injured: bool, blocking: bool, contacts: list[dict]) -> dict:
108     if not contacts:
109         return {"contacts": [], "reason": "No nearby services found"}
110
111     api_key = os.getenv("GEMINI_API_KEY", "")
112     if not api_key:
113         logger.info("GEMINI_API_KEY not set [U+2014] using rule-based triage")
114         return rule_based_triage(injured, blocking, contacts)
115
116     if not gemini_quota.can_call():
117         logger.info("Gemini quota guard triggered [U+2014] using rule-based triage")
118         return rule_based_triage(injured, blocking, contacts)
119
120     try:
121         situation = []
122         if injured:
123             situation.append("PEOPLE ARE INJURED and need medical attention")
124         if blocking:
125             situation.append("THE VEHICLE IS BLOCKING THE ROAD")
126         if not situation:
127             situation.append("no injuries, vehicle may need roadside assistance")
128         situation_text = "; ".join(situation)
129
130         summary = [
131             {
132                 "name": c.get("name", "?"),
133                 "category": c.get("category", "?"),
134                 "distance_km": c.get("distance", "?"),
135             }
136             for c in contacts
137         ]
138
139         user_message = (
140             f"SITUATION: {situation_text}.\n\n"
141             f"Services found nearby (currently sorted by distance only):\n"
142             f"{json.dumps(summary, indent=2)}\n\n"
143             f"Full contact data to reorder (return ALL of these, "
144             f"just in the priority order you decide):\n"
145             f"{json.dumps(contacts, indent=2)}"
146         )
147
148     # Gemini puts the system prompt in 'systemInstruction', not in messages.
149     # JSON mode is opt-in via responseMimeType.
150     payload = {
151         "systemInstruction": {"parts": [{"text": SYSTEM_PROMPT}]},
152         "contents": [
153             {"role": "user", "parts": [{"text": user_message}]},
154         ],
155         "generationConfig": {
156             "temperature": 0.2,

```

```

157         "maxOutputTokens": MAX_TOKENS,
158         "responseMimeType": "application/json",
159         "thinkingConfig": {"thinkingBudget": 0},
160     },
161 }
162
163 url = GEMINI_URL.format(model=MODEL, key=api_key)
164 async with httpx.AsyncClient(timeout=TIMEOUT_S) as client:
165     r = await client.post(url, json=payload)
166     r.raise_for_status()
167     response_json = r.json()
168     await gemini_quota.record_call()
169
170 raw = extract_gemini_text(response_json)
171 if not raw:
172     logger.warning("AI returned empty response [U+00B7] using rule-based fallback")
173     return rule_based_triage(injured, blocking, contacts)
174
175 cleaned = _FENCE_RE.sub("", raw).strip()
176
177 try:
178     parsed = json.loads(cleaned)
179 except json.JSONDecodeError:
180     logger.warning("AI returned non-JSON [U+00B7] using rule-based fallback")
181     return rule_based_triage(injured, blocking, contacts)
182
183 validated = _validate_ai_result(parsed, len(contacts))
184 if validated is None:
185     logger.warning("AI returned malformed response [U+00B7] using rule-based
↳ fallback")
186     return rule_based_triage(injured, blocking, contacts)
187
188     return validated
189
190 except Exception as exc:
191     logger.warning(f"AI triage call failed ({type(exc).__name__}) [U+00B7] using rule-
↳ based fallback")
192     return rule_based_triage(injured, blocking, contacts)

```


6.3 Source index

Files included verbatim: **2**.

Document version. Generated from `main`. Re-run `python scripts/build_submission_tex.py` after any code change to regenerate.